

A probability experiment

A probability experiment produces an outcome that is uncertain before it occurs.

What are the possible outcomes when a fair six-sided die is rolled?

Sample space

The sample space contains all possible outcomes.

What sample space describes two ordered tosses of a fair coin?

An event

An event is a set of outcomes from the sample space.

For a die roll, what outcomes belong to “an even number”?

Equally likely outcomes

If all outcomes are equally likely, event probability is favorable outcomes divided by total outcomes.

What is the probability of a value greater than 4 on a fair six-sided die?

Unequal probabilities

A spinner has a red region covering half the circle and two other regions covering one quarter each.

Are its three color outcomes equally likely?

Probability bounds

An event probability lies between 0 and 1.

What do probabilities 0 and 1 mean?

Relative frequency

Synthetic simulation: 124 heads in 250 independent fair-coin tosses.

What is the observed relative frequency of heads?

Long-run behavior

A fair coin has probability 0.5 of heads on each independent toss.

Must the first ten tosses include exactly five heads?

A complement

The complement of A contains outcomes outside A.

$$P(A^c) = 1 - P(A)$$

If $P(A) = 0.35$, what is $P(A\text{-complement})$?

Translate “not”

A fair six-sided die is rolled. A is “roll a 6.”

What outcomes belong to A-complement?

The intersection

A AND B means both events occur.

For a fair die, A = even and B = greater than 3. Which outcomes belong to A AND B?

The union

A OR B means A occurs, B occurs, or both occur.

For a fair die, $A = \text{even}$ and $B = \text{greater than 3}$. Which outcomes belong to A OR B?

Inclusive “or”

One card is drawn from a standard 52-card deck.

Does “a heart or a king” include the king of hearts?

Conditional probability

$P(A | B)$ is the probability of A when the outcome is restricted to B.

For a fair die, what is $P(\text{even} | \text{greater than } 3)$?

The conditioned group

Synthetic survey: 20 students use tablets; 8 of those also use laptops. Select one student uniformly from the tablet users.

What is the denominator for $P(\text{laptop} | \text{tablet})$?

Reverse the condition

Synthetic survey: 20 tablet users, 30 laptop users, and 8 use both. Select one of these surveyed students uniformly.

Is $P(\text{laptop} \mid \text{tablet})$ equal to $P(\text{tablet} \mid \text{laptop})$?

Without replacement

A box holds 3 red and 2 blue tokens. Draw one token, keep it, then draw another.

How does the first draw affect the possible second draw?

Write a probability event

Let X be the number of heads in three independent fair-coin tosses.

How would you write “at least two heads” using X ?